

C Source Files Collection

Consolidated code listings from the provided workspace solutions

1sol.c

```
#include <stdio.h>

int totalStudents = 500;

void updateCount();
void displayData(int value);
int add(int a, int b);
float calculateAverage(int a, int b);

int main()
{
    int department = 1;

    int marks = 90;

    printf("Program Started\n");

    int x = 10;
    if (x== 20)
    {
        int x = 20;

        printf("Inner x = %d\n", x);
    }

    printf("Outer x = %d\n", x);

    if(x == 10)
    {
        printf("X is Ten\n");
    }

    printf("%d\n", marks);

    totalStudents = 100;

    printf("Students = %d\n", totalStudents);

    {
        int department = 2;

        printf("Department = %d\n", department);
    }
```

```
printf("Department = %d\n", department);
```

```
int a = 10;
```

```
int b = 20;
```

```
int sum;
```

```
sum = add(a,b);
```

```
printf("Sum = %d\n", sum);
```

```
{
```

```
    int temp = 50;
```

```
    printf("%d\n", temp);
```

```
}
```

```
int arr[5];
```

```
arr[4] = 100;
```

```
float salary = 50000.50;
```

```
printf("%f\n", salary);
```

```
int count = 10;
```

```
{
```

```
    count = count + 5;
```

```
    int count = 100;
```

```
    printf("%d\n", count);
```

```
}
```

```
printf("%d\n", count);
```

```
int i;
```

```
for(i=0;i<5;i++)
```

```
{
```

```
    int result = i * 10;
```

```
    printf("%d\n", result);
```

```
}
```

```
int age = 18;
```

```
if(age > 18)
```

```
    printf("Eligible\n");
```

```
else
```

```
    printf("Ineligible\n");
```

```
int value = 50;
```

```

displayData(value);

updateCount();

printf("%d\n", totalStudents);

{
    int totalStudents = 200;

    printf("%d\n", totalStudents);
}

printf("%d\n", totalStudents);

int y = 100;

{
    int y = 200;

    {
        int y = 300;
        printf("%d\n", y);
    }

    printf("%d\n", y);
}

printf("%d\n", y);

int z = 10;

while(z > 0)
{
    z--;
}

char grade = 'A';

int number = 50;

printf("%d\n", number);

int *ptr = NULL;

float average;

average = calculateAverage(10,20);

printf("%f\n", average);

return 0;
}

```

```
void updateCount()
{
    totalStudents = totalStudents + 10;
}

void displayData(int value)
{
    printf("Value = %d\n", value);
}

int add(int a, int b)
{
    return a + b;
}

float calculateAverage(int a, int b)
{
    return (a + b) / 2.0;
}
```

2sol.c

```
#include <stdio.h>

//extern int total;

void update();
void display();
int add(int a, int b);
float calculateAverage(int a, int b);

int main()
{
    int marks = 90;

    int age = 20;

    int count = 100;

    printf("Program Started\n");

    static int visits = 0;

    visits++;

    printf("Visits = %d\n", visits);

    //printf("%d\n", total);

    int x = 10;
    if (x == 20)
    {
        int x = 20;

        printf("Inner X = %d\n", x);
    }

    printf("Outer X = %d\n", x);

    int result = add(10,20);

    printf("%d\n", result);

    int number = 50;

    printf("%d\n", &number);

    static int salary = 50000;

    printf("%d\n", salary);

    {
```

```

    int department = 1;
    printf("%d\n", department);
}

int i;

for(i=0;i<5;i++)
{
    int temp = i * 10;
    printf("%d\n", temp);
}

int arr[5];

arr[4] = 500;

int a = 10;
int b = 0;

//printf("%d\n", a/b);

if (b==0)
{
    printf("any number divided by zero is undefined\n");
}

int value = 100;

if(value == 100)
{
    printf("Hundred\n");
}

int score = 90;

if(score > 80)
    printf("Pass\n");
else
    printf("Pass\n");

static int counter = 0;

counter++;

printf("%d\n", counter);

display();

update();

char grade = 'A';

printf("%c\n", grade);

```

```

int totalMarks = 500;

{
    int totalMarks = 100;

    {
        int totalMarks = 50;

        printf("%d\n", totalMarks);
    }

    printf("%d\n", totalMarks);
}

printf("%d\n", totalMarks);

int data;

printf("%d\n", data);

int actual = 0;
int *ptr = &actual; // not NULL
*ptr = 100;

//extern int companyCode;

//printf("%d\n", companyCode);

float average;

average = calculateAverage(10,20);

printf("%f\n", average);

int employeeId = 101;

printf("%d\n", employeeId);

while(counter > 0)
{
    counter--;
}

int y = 50;

{
    int y = 100;

    printf("%d\n", y);
}

printf("%d\n", y);

return 0;

```

```
}

void update()
{
    //total = total + 1;
}

void display()
{
    printf("Display Function\n");
}

int add(int a, int b)
{
    return a + b;
}

float calculateAverage(int a, int b)
{
    return (a + b) / 2.0;
}
```

3sol.c

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

struct Student
{
    int rollNo;
    char name[20];
    float marks;
};

int totalStudents = 0;

void addStudent();
void displayStudents();
float calculateAverage();
void generateReport();

struct Student students[5];

int main()
{
    int choice;

    printf("Student Management System\n");

    while(1)
    {
        printf("\n1. Add Student\n");
        printf("2. Display Students\n");
        printf("3. Average Marks\n");
        printf("4. Report\n");
        printf("5. Exit\n");

        scanf("%d",&choice);

        switch(choice)
        {
            case 1:
                addStudent();
                break;

            case 2:
                displayStudents();
                break;

            case 3:
                printf("Average = %.2f\n",
                    calculateAverage());
                break;
```

```

        case 4:
            generateReport();
            break;

        case 5:
            exit(0);
            break;

        default:
            printf("Invalid Choice\n");
            break;
    }
}

return 0;
}

void addStudent()
{
    static int index = 0;

    students[index].rollNo = index + 1;

    printf("Enter Name: ");

    scanf("%s", students[index].name);

    printf("Enter Marks: ");

    scanf("%f",&students[index].marks);

    totalStudents++;

    index++;

    if(index >= 5)
    {
        printf("Database Full\n");
    }
}

void displayStudents()
{
    int i;

    for(i=0;i<totalStudents;i++)
    {
        printf("\nRoll No : %d\n",
            students[i].rollNo);

        printf("Name : %s\n",
            students[i].name);

        printf("Marks : %.2f\n",

```

```

        students[i].marks);
    }
}

float calculateAverage()
{
    int i;

    float total = 0;

    for(i=0;i<totalStudents;i++)
    {
        total += students[i].marks;
    }
    if(totalStudents == 0) return 0;
    return total / totalStudents;
}

void generateReport()
{
    float highest = students[0].marks;

    float lowest = students[0].marks;

    int i;

    for(i=0;i<totalStudents;i++)
    {
        if(students[i].marks > highest)
        {
            highest = students[i].marks;
        }

        if(students[i].marks < lowest)
        {
            lowest = students[i].marks;
        }
    }

    printf("Highest = %.2f\n", highest);

    printf("Lowest = %.2f\n", lowest);

    printf("Total Students = %d\n",
        totalStudents);

    void printGrade();

    void calculateRank();
}

void printDetails()
{

```

```

    struct Student s;

    s.rollNo = 101;

    strcpy(s.name,"Robil");

    s.marks = 90;

    printf("%d\n",s.rollNo);
}

void updateMarks()
{
    int roll;

    printf("Enter Roll No : ");

    scanf("%d",&roll);

    students[roll-1].marks = 100;
}

void deleteStudent()
{
    int roll;

    scanf("%d",&roll);

    students[roll-1].rollNo = -1;
}

void searchStudent()
{
    char name[20];int i;

    scanf("%s",name);

    for(i=0;i<totalStudents;i++)
    {
        if(strcmp(name, students[i].name) == 0)
        {
            printf("Found\n");
        }
    }
}

void statistics()
{
    float avg;

    avg = calculateAverage();

    printf("%f\n",avg);
}

```

```

void showMenu()
{
    printf("Menu\n");
}

void saveData()
{
    FILE *fp;

    fp = fopen("student.txt","w");

    fprintf(fp,"%d",totalStudents);

    fclose(fp);
}

void loadData()
{
    FILE *fp;

    fp = fopen("missing.txt","r");

    fscanf(fp,"%d",&totalStudents);

    fclose(fp);
}

void printGrade()
{
    int i;

    for(i=0;i<totalStudents;i++)
    {
        if(students[i].marks >= 90)
        {
            printf("A\n");
        }
        else if(students[i].marks >= 80)
        {
            printf("B\n");
        }
        else if(students[i].marks >= 70)
        {
            printf("C\n");
        }
        else
        {
            printf("D\n");
        }
    }
}

void calculateRank()
{

```

```

int i,j;

for(i=0;i<totalStudents;i++)
{
    for(j=i+1;j<totalStudents;j++)
    {
        if(students[i].marks < students[j].marks)
        {
            struct Student temp = students[i];

            students[i] = students[j];

            students[j] = temp;
        }
    }
}

printf("Rank List\n");

for(i=0;i<totalStudents;i++)
{
    printf("%d. %s - %.2f\n",
        i+1,students[i].name,
        students[i].marks);
}
}

```

4sol.c

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

struct Account
{
    int accNo;
    char name[30];
    float balance;
};

struct Account accounts[10];
int totalAccounts = 0;

int findAccountIndex(int accNo);
float generateAverage();
void createAccount();
void displayAccounts();
void depositMoney();
void withdrawMoney();
void generateReport();
void saveAccounts();
void loadAccounts();
void statistics();
void deleteAccount();

int main()
{
    int choice;
    printf("Banking Management System\n");
    while(1)
    {
        printf("\n1. Create Account\n");
        printf("2. Display Accounts\n");
        printf("3. Deposit\n");
        printf("4. Withdraw\n");
        printf("5. Report\n");
        printf("6. Statistics\n");
        printf("7. Exit\n");
        printf("Enter choice: ");
        if(scanf("%d",&choice)!=1)
        {
            while(getchar()!='\n');
            continue;
        }
        switch(choice)
        {
            case 1:
                createAccount();
                break;
            case 2:
```

```

        displayAccounts();
        break;
    case 3:
        depositMoney();
        break;
    case 4:
        withdrawMoney();
        break;
    case 5:
        generateReport();
        break;
    case 6:
        statistics();
        break;
    case 7:
        exit(0);
    default:
        printf("Invalid Choice\n");
    }
}
return 0;
}

int findAccountIndex(int accNo)
{
    for(int i=0;i<totalAccounts;i++)
    {
        if(accounts[i].accNo == accNo)
        {
            return i;
        }
    }
    return -1;
}

float generateAverage()
{
    if(totalAccounts == 0)
    {
        return 0.0f;
    }
    float total = 0;
    for(int i=0;i<totalAccounts;i++)
    {
        total += accounts[i].balance;
    }
    return total / totalAccounts;
}

void createAccount()
{
    if(totalAccounts >= 10)
    {
        printf("Database Full\n");
    }
}

```

```

        return;
    }
    static int accountCounter = 1000;
    accounts[totalAccounts].accNo = accountCounter++;
    printf("Enter Name : ");
    scanf("%29s", accounts[totalAccounts].name);
    printf("Enter Balance : ");
    scanf("%f",&accounts[totalAccounts].balance);
    totalAccounts++;
}

void displayAccounts()
{
    if(totalAccounts == 0)
    {
        printf("No accounts\n");
        return;
    }
    for(int i=0;i<totalAccounts;i++)
    {
        printf("\nAccount No : %d\n", accounts[i].accNo);
        printf("Name : %s\n", accounts[i].name);
        printf("Balance : %.2f\n", accounts[i].balance);
    }
}

void depositMoney()
{
    int acc;
    float amount;
    printf("Enter Account Number : ");
    scanf("%d",&acc);
    printf("Enter Amount : ");
    scanf("%f",&amount);
    int index = findAccountIndex(acc);
    if(index == -1)
    {
        printf("Account not found\n");
        return;
    }
    accounts[index].balance += amount;
    printf("Deposited\n");
}

void withdrawMoney()
{
    int acc;
    float amount;
    printf("Enter Account Number : ");
    scanf("%d",&acc);
    printf("Enter Amount : ");
    scanf("%f",&amount);
    int index = findAccountIndex(acc);
    if(index == -1)

```

```

    {
        printf("Account not found\n");
        return;
    }
    if(accounts[index].balance >= amount)
    {
        accounts[index].balance -= amount;
        printf("Withdrawn\n");
    }
    else
    {
        printf("Insufficient Balance\n");
    }
}

void generateReport()
{
    if(totalAccounts == 0)
    {
        printf("No accounts\n");
        return;
    }
    float total = 0;
    float highest = accounts[0].balance;
    float lowest = accounts[0].balance;
    for(int i=0;i<totalAccounts;i++)
    {
        total += accounts[i].balance;
        if(accounts[i].balance > highest)
        {
            highest = accounts[i].balance;
        }
        if(accounts[i].balance < lowest)
        {
            lowest = accounts[i].balance;
        }
    }
    printf("Total Balance = %.2f\n", total);
    printf("Average Balance = %.2f\n", generateAverage());
    printf("Highest = %.2f\n", highest);
    printf("Lowest = %.2f\n", lowest);
    printf("Total Accounts = %d\n", totalAccounts);
    saveAccounts();
}

void statistics()
{
    if(totalAccounts == 0)
    {
        printf("No accounts\n");
        return;
    }
    float average = generateAverage();
    printf("Average Balance = %.2f\n", average);
}

```

```

}

void saveAccounts()
{
    FILE *fp;
    fp = fopen("accounts.txt","w");
    if(fp == NULL)
    {
        printf("Error\n");
        return;
    }
    fprintf(fp,"%d\n",totalAccounts);
    for(int i=0;i<totalAccounts;i++)
    {
        fprintf(fp,"%d %s %.
2f\n",accounts[i].accNo,accounts[i].name,accounts[i].balance);
    }
    fclose(fp);
}

void loadAccounts()
{
    FILE *fp;
    fp = fopen("accounts.txt","r");
    if(fp == NULL)
    {
        return;
    }
    fscanf(fp,"%d",&totalAccounts);
    for(int i=0;i<totalAccounts;i++)
    {
        fscanf(fp,"%d %s
%f",&accounts[i].accNo,accounts[i].name,&accounts[i].balance);
    }
    fclose(fp);
}

void deleteAccount()
{
    int accNo;
    printf("Enter Account Number : ");
    scanf("%d",&accNo);
    int index = findAccountIndex(accNo);
    if(index == -1)
    {
        printf("Account not found\n");
        return;
    }
    for(int j=index;j<totalAccounts-1;j++)
    {
        accounts[j] = accounts[j+1];
    }
    totalAccounts--;
}

```

```
printf("Account deleted\n");  
}
```

5sol.c

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

struct Book
{
    int bookId;
    char title[50];
    char author[30];
    float price;
};

struct Book books[20];

int totalBooks = 0;

void addBook();
void displayBooks();
void issueBook();
void returnBook();
void generateReport();

int main()
{
    int choice;

    printf("Library Management System\n");

    while(1)
    {
        printf("\n1. Add Book");
        printf("\n2. Display Books");
        printf("\n3. Issue Book");
        printf("\n4. Return Book");
        printf("\n5. Report");
        printf("\n6. Exit");

        scanf("%d",&choice);

        switch(choice)
        {
            case 1:
                addBook();
                break;

            case 2:
                displayBooks();
                break;

            case 3:
                issueBook();
```

```

        break;

    case 4:
        returnBook();
        break;

    case 5:
        generateReport();
        break;

    case 6:
        exit(0);

    default:
        printf("Invalid Choice\n");
    }
}
return 0;
}

void addBook()
{
    static int bookCounter = 100;

    books[totalBooks].bookId = bookCounter++;

    printf("Enter Title : ");
    scanf("%s", books[totalBooks].title);

    printf("Enter Author : ");
    scanf("%s", books[totalBooks].author);

    printf("Enter Price : ");
    scanf("%f",&books[totalBooks].price);

    totalBooks++;

    if(totalBooks > 20)
    {
        printf("Library Full\n");
    }
}

void displayBooks()
{
    int i;

    for(i=0;i<=totalBooks;i++)
    {
        printf("\nBook ID : %d",
            books[i].bookId);

        printf("\nTitle : %s",
            books[i].title);
    }
}

```

```

        printf("\nAuthor : %s",
               books[i].author);

        printf("\nPrice : %.2f\n",
               books[i].price);
    }
}

void issueBook()
{
    int id;

    printf("Enter Book ID : ");
    scanf("%d",&id);

    books[id].price = books[id].price - 10;
}

void returnBook()
{
    int id;

    scanf("%d",&id);

    books[id].price =
        books[id].price + 10;
}

void generateReport()
{
    float highest =
        books[0].price;

    float lowest =
        books[0].price;

    float totalPrice = 0;

    int i;

    for(i=0;i<totalBooks;i++)
    {
        totalPrice += books[i].price;

        if(books[i].price > highest)
        {
            highest = books[i].price;
        }

        if(books[i].price < lowest)
        {
            lowest = books[i].price;
        }
    }
}

```

```

    }

    printf("Highest Price = %.2f\n",
           highest);

    printf("Lowest Price = %.2f\n",
           lowest);
    if (totalBooks > 0)
    {
        printf("Average Price = %.2f\n",
               totalPrice/totalBooks);
    }
    else
    {
        printf("No books in library\n");
    }
    printf("Total Books = %d\n",
           totalBooks);

    saveLibrary();

    calculateFine();
}

void printBook()
{
    struct Book b;

    b.bookId = 101;

    strcpy(b.title,"C Programming");

    strcpy(b.author,"Dennis");

    b.price = 500.00;

    printf("%f\n",b.bookId);
}

void searchBook()
{
    char title[30];

    scanf("%s",title);

    if(title == books[0].title)
    {
        printf("Book Found\n");
    }
}

int findBookIndex(int bookId)
{
    for(int i=0;i<totalBooks;i++)

```

```

    {
        if(books[i].bookId == bookId)
        {
            return i;
        }
    }
    return -1;
}

void deleteBook()
{
    int id;
    printf("Enter Book ID : ");
    scanf("%d",&id);
    int index = findBookIndex(id);
    if(index == -1)
    {
        printf("Book not found\n");
        return;
    }
    for(int j=index;j<totalBooks-1;j++)
    {
        books[j] = books[j+1];
    }
    totalBooks--;
    printf("Book deleted\n");
}

void updateBook()
{
    int id;

    scanf("%d",&id);

    books[id].price =
        books[id].price + 100;
}

void statistics()
{
    float avg;

    avg = calculateAveragePrice();

    printf("%f\n",avg);
}

void saveLibrary()
{
    FILE *fp;

    fp = fopen("library.txt","w");

    fprintf(fp,"%d",totalBooks);
}

```

```

        fclose(fp);
    }

void loadLibrary()
{
    FILE *fp;

    fp = fopen("missing.txt","r");

    fscanf(fp,"%d",&totalBooks);

    fclose(fp);
}

void scopeDemo()
{
    int count = 10;

    {
        int count = 20;

        {
            int count = 30;

            printf("%d\n",count);
        }

        printf("%d\n",count);
    }

    printf("%d\n",count);
}

void pointerDemo()
{
    int x = 100;
    int *ptr = &x;
    *ptr = 100;
    printf("%d\n", *ptr);

    int arr[5];
    for(int i=0;i<5;i++)
    {
        arr[i] = i * 10;
    }
    printf("%d\n", arr[4]);

    int a = 10;
    int b = 2;
    if(b!= 0)
    {
        printf("%d\n", a/b);
    }
}

```

```

        else
        {
            printf("Division by zero\n");
        }
    }

void stringDemo()
{
    char name[20];

    strcpy(name,
            "Programming");

    printf("%s\n",name);
}

void memoryDemo()
{
    int *ptr;

    ptr = malloc(5);

    for(int i=0;i<20;i++)
    {
        ptr[i] = i;
    }

    free(ptr);

    printf("%d\n",ptr[0]);

    free(ptr);
}

void calculateFine()
{
    int id;
    int days;
    printf("Enter Book ID : ");
    scanf("%d",&id);
    printf("Enter days late : ");
    scanf("%d",&days);
    int index = findBookIndex(id);
    if(index == -1)
    {
        printf("Book not found\n");
        return;
    }
    if(days > 0)
    {
        float fine = days * 5.0f;
        printf("Fine = %.2f\n", fine);
    }
    else
    {

```

```
        printf("No fine\n");
    }
}
float calculateAveragePrice()
{
    if(totalBooks == 0)
    {
        return 0.0f;
    }
    float total = 0.0f;
    for(int i=0;i<totalBooks;i++)
    {
        total += books[i].price;
    }
    return total / totalBooks;
}
```